Monetary Policy Transmission in Peru: A Systematic Comparison of Identification Strategies

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Abstract

We estimate the effects of monetary policy on GDP and poverty in Peru using quarterly data from 2004Q2 to 2025Q3 ($T = 85$). We systematically compare seven identification strategies—recursive (Cholesky), sign restrictions, narrative sign restrictions, local projections, two Proxy-SVAR approaches using interbank rate surprises and AI-classified BCRP communication tone (325 *Notas Informativas*, 2001–2026), and a GDP–poverty reduced-form regression—and find that only recursive identification produces interpretable estimates. The Cholesky VAR(1) yields a peak GDP response of -0.195 pp per 100 bp rate hike (90% bootstrap CI: $[-0.698, +0.271]$), consistent with the Peru SVAR literature (-0.25 to -0.30 pp). Modern identification methods fail because Peru's interbank rate is administered, no rate-futures market exists for policy surprise extraction, and the monetary record since 2003 provides too few sharp episodes for narrative restrictions. Our most robust result is a GDP–poverty elasticity of $\hat{\beta} = -0.656$ ($t = -5.69$, $R^2 = 0.67$, $N = 18$), estimated from ENAHO annual data 2005–2024 and stable across sub-periods. Chaining the two elasticities implies a 100 bp hike may raise poverty by approximately $+0.13$ pp, though this inherits the large uncertainty of the rate–GDP link. The paper explains why the Peru SVAR literature relies on recursive identification and cautions against applying Proxy-SVAR and narrative methods—developed for the US and Euro Area—to emerging markets with administered policy rates.

JEL codes: E52, E32, I32, C32.

Keywords: monetary policy transmission, SVAR, local projections, narrative identification, Proxy-SVAR, poverty, Peru.

1 Introduction

Understanding how monetary policy transmits to real activity is a central question in macroeconomics. The global literature has developed an expanding identification toolkit: recursive (Cholesky) ordering, sign restrictions (Uhlig, 2005), narrative restrictions (Antolín-Díaz and Rubio-Ramírez, 2018), and external-instrument Proxy-SVARs (Stock and Watson, 2018; Gertler and Karadi, 2015; Montiel Olea et al., 2021). These methods have been applied extensively to the United States and Euro Area, and increasingly to advanced economies such as Australia. Their applicability to small open emerging-market economies (EMEs) with institutional features that differ markedly from the US—administered policy rates, no OIS market, limited monetary history—has received far less systematic attention.

Peru offers a particularly instructive case. The Banco Central de Reserva del Perú (BCRP) has operated a credible inflation-targeting framework since 2002, with a target of $2\% \pm 1$ pp. Yet Peru's monetary implementation differs fundamentally from the US Federal Reserve: the interbank overnight rate is *administered*—the BCRP ensures the interbank rate equals the reference rate by construction, leaving no market-determined component from which to extract policy surprises. This single institutional feature, combined with the absence of rate-futures markets and a relatively uneventful monetary history, renders the modern identification toolkit mostly inapplicable.

The existing Peru-specific literature—Pérez Rojo and Rodríguez (2024), Castillo et al. (2011), and Portilla et al. (2022)—uniformly relies on recursive identification without documenting why more modern alternatives fail. This paper fills that gap with four contributions:

1. **First systematic comparison** of seven identification strategies for monetary policy in Peru within a single unified dataset ($T = 85$, 2004Q2–2025Q3; see Table 12).

2. **First Proxy-SVAR with BCRP communication analysis:** we classify 325 *Notas Informativas del Programa Monetario* (2001–2026) using both a hawkish/dovish dictionary and a large language model (Claude Haiku, -100 to $+100$ scale), yielding the

most comprehensive BCRP tone dataset to date (Table 2).

3. **A well-identified GDP–poverty elasticity** ($\hat{\beta} = -0.656$, $p < 0.001$, $N = 18$; Table 10) that chains monetary policy effects to poverty outcomes, stable across sub-periods (Table 11).

4. **An explanation of why modern identification fails** in Peru's institutional environment, contributing to the emerging literature on monetary policy transmission in EMEs (Ha et al., 2025).

The Cholesky benchmark (Figure 3) yields a peak GDP response of -0.195 pp per 100 bp consistent with published Peru estimates. Sign restrictions produce an identified set too wide for inference. Narrative restrictions are blocked by a fundamental COVID-dummy–narrative tension. Proxy-SVAR with the interbank rate fails relevance ($F = 4.73$; Table 8), and Proxy-SVAR with tone passes relevance only when the instrument retains anticipated policy (violating exogeneity; Table 9).

Section 2 describes Peru's monetary framework. Section 3 presents the data. Section 4 develops the econometric framework. Section 5 reports results. Section 6 presents the master comparison. Section 7 discusses implications. Section 8 concludes.

2 Institutional Background

2.1 BCRP Monetary Policy Framework

The BCRP adopted inflation targeting in 2002, with a target band of $2\% \pm 1$ pp. The reference rate is announced monthly on a pre-set calendar. Unlike the Federal Reserve, the BCRP keeps the interbank overnight rate equal to the reference rate through active liquidity provision or absorption, making it administered by construction.

Additional distinguishing features include: (i) active foreign exchange intervention in a managed float; (ii) partial dollarization, with significant deposits and credit in both Peruvian

soles (PEN) and US dollars; and (iii) the absence of any OIS or rate-futures market for PEN, precluding high-frequency extraction of market rate expectations.

2.2 Why Peru Differs from the US and Euro Area

Three features make modern identification strategies particularly difficult.

Administered interbank rate. The interbank rate equals the reference rate by BCRP design, leaving no "surprise" component in the sense of Gertler and Karadi (2015). The interbank "surprise" used as a Proxy-SVAR instrument is mechanically collinear with the reference rate change already in the VAR (Section 5.5).

No rate expectations market. Without OIS or futures contracts on PEN rates, it is impossible to separate the expected from the unexpected component of a rate decision, as is standard in the US literature (Miranda-Agrippino and Ricco, 2021).

Limited sharp monetary episodes. Since 2002, BCRP changes have been small and gradual, with only two exceptions: the emergency COVID cuts of 2020 (2.25% to 0.25%) and the 2021–2022 hiking cycle (0.25% to 7.75%). As Read and Zubairy (2023) demonstrate, even the US narrative identification literature struggles with fewer than a handful of clear episodes; two episodes are insufficient for Peru.

2.3 BCRP Reference Rate History

Figure 1 shows the BCRP reference rate from 2003 to 2025Q3, with key episodes annotated: the GFC cutting cycle (2008–2009, from 6.5% to 1.25%), the 2010–2011 hiking cycle (to 4.25%), gradual cuts in 2014–2015, the COVID emergency cuts to 0.25% in 2020, the aggressive 2021–2022 hiking cycle to 7.75%, and subsequent easing to 4.25% by end-2025.

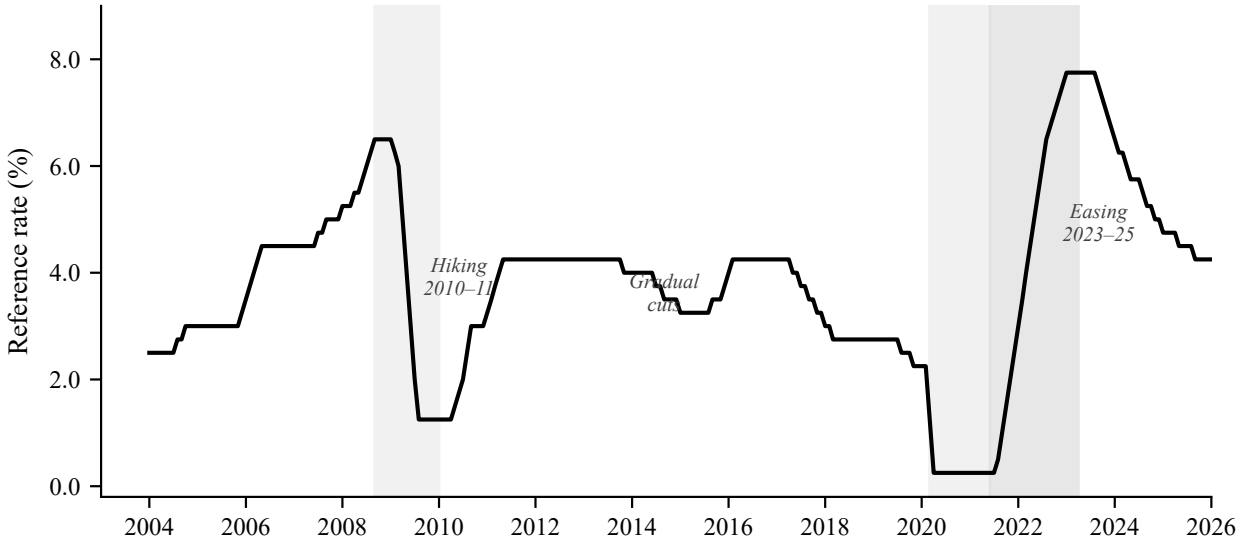

Figure 1: BCRP Reference Rate, 2003–2025. Key episodes: GFC cuts (2008–2009), 2010–2011 hiking cycle, COVID emergency cuts to 0.25% (2020), aggressive hiking to 7.75% (2021–2022), and gradual easing (2023–2025). Source: BCRP series PD04722MM.

3 Data

3.1 Macroeconomic Variables

The VAR uses five quarterly series from the BCRP data portal: terms of trade (PN38923BM), real GDP quarter-on-quarter growth (PN01731AM, summed from monthly seasonally adjusted index), CPI quarter-on-quarter change (PN01271PM), PEN/USD exchange rate change (PN01246PM), and BCRP reference rate change (PD04722MM). The sample runs from 2004Q2 to 2025Q3 ($T = 85$). Summary statistics are in Table 1; all series are stationary by ADF tests ($p < 0.001$ for each).

COVID treatment follows the Frisch-Waugh-Lovell (FWL) principle: indicator variables for 2020Q1 and 2020Q2 are partialled out of all series before estimation. This is preferred to a contemporaneous dummy in the VAR because a dummy in Y_t but not the lags creates inconsistency in VAR($p \geq 2$) systems by contaminating lagged regressors.

Table 1: Summary Statistics, 2004Q2–2025Q3

Variable	Mean	SD	Min	Max	ADF stat	ADF p
GDP (qoq, pp)	0.013	2.810	-10.890	4.851	-4.80	0.000
CPI (qoq, pp)	0.527	0.574	-0.680	2.510	-5.66	0.000
FX (qoq, %Δ)	0.085	2.104	-6.502	5.900	-6.50	0.000
Rate (Δ pp)	0.009	0.503	-1.250	1.250	-5.90	0.000
ToT (qoq, pp)	0.170	4.050	-16.010	9.220	-7.51	0.000

Sample: 2004Q2–2025Q3 ($T = 85$). ADF with intercept; all series stationary.
COVID FWL: 2020Q1–Q2 partialled out in estimation.

3.2 BCRP Communications Corpus

We construct a dataset of 325 *Notas Informativas del Programa Monetario*—the official monthly communications explaining each rate decision—covering 2001 to 2026. Documents were downloaded from the BCRP website and the Wayback Machine; text was extracted with `pdfplumber`.

Each note is classified by tone using two methods: (a) a dictionary-based approach using a hawkish/dovish lexicon adapted from Lahura and Vega (2020), producing scores in $[-1, +1]$; and (b) an LLM-based approach using Claude Haiku, prompted to score each note on $[-100, +100]$. Table 2 presents classification statistics. The correlation between dictionary and LLM scores is 0.373, indicating moderate agreement. Both methods correctly flag 2009 as dovish and 2022 as hawkish. Figure 2 shows the tone series over time.

3.3 Poverty Data

The poverty measure is the official annual monetary poverty headcount ratio from INEI/ENAHO household surveys, 2005–2024. COVID years 2020–2021 are excluded due to lockdown disruptions and INEI methodology changes. The regression sample contains $N = 18$ annual observations. Table 4 lists the full dataset.

Table 2: BCRP Communication Tone: Summary Statistics

Statistic	Dictionary	LLM (Claude Haiku)
Corpus size (notes)	325	
Date range	2001–2026	
Mean score	-0.062	-7.7
SD	0.512	40.2
Min	-1.0	-85
Max	1.0	75
Correlation (dict vs LLM)	0.373	

Dictionary: hawkish/dovish lexicon adapted from Lahura & Vega (2020 CEMLA). LLM: Claude Haiku (`claude-haiku-4-5-20251001`), prompt scored each Nota Informativa on $[-100, +100]$ with chain-of-thought rationale.

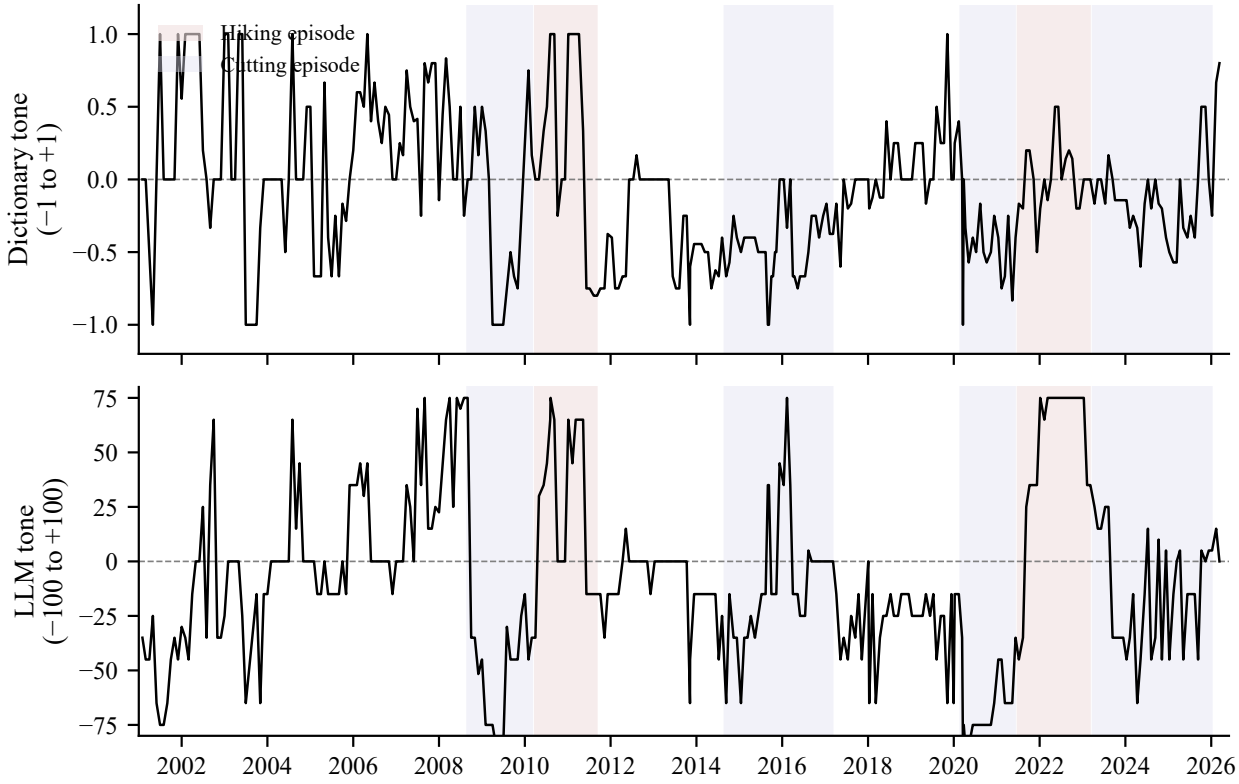

Figure 2: BCRP Communication Tone, 2001–2026. Bars: dictionary score (left axis). Line: LLM score (right axis). Both flag 2009 (GFC emergency cuts) as strongly dovish and 2022 (hiking cycle) as hawkish. Dictionary–LLM correlation: $r = 0.373$.

Table 3: BCRP Communication Tone by Year, 2001–2026

Year	N	Dict. mean	(SD)	LLM mean	(SD)
2001	12	+0.00	(0.60)	-50.0	(16.2)
2002	13	+0.46	(0.53)	-10.4	(34.2)
2003	10	+0.17	(0.81)	-21.0	(25.3)
2004	12	+0.08	(0.36)	+9.2	(22.7)
2005	12	-0.27	(0.46)	-5.8	(14.7)
2006	13	+0.47	(0.32)	+15.8	(21.1)
2007	14	+0.47	(0.34)	+31.4	(29.6)
2008	17	+0.22	(0.51)	+27.6	(53.2)
2009	13	-0.53	(0.53)	-57.3	(32.2)
2010	15	+0.32	(0.57)	+12.0	(40.5)
2011	12	-0.05	(0.84)	+15.0	(41.3)
2012	12	-0.26	(0.36)	-3.8	(9.3)
2013	13	-0.36	(0.36)	-9.6	(21.0)
2014	12	-0.52	(0.14)	-25.8	(16.2)
2015	15	-0.53	(0.24)	-13.0	(29.8)
2016	15	-0.45	(0.26)	+0.3	(28.4)
2017	12	-0.17	(0.19)	-22.9	(17.9)
2018	13	+0.03	(0.17)	-26.2	(19.2)
2019	13	+0.20	(0.30)	-30.4	(17.1)
2020	14	-0.30	(0.37)	-65.0	(24.5)
2021	12	-0.31	(0.34)	-22.5	(42.0)
2022	12	+0.06	(0.25)	+74.2	(2.9)
2023	12	-0.05	(0.10)	+9.2	(36.0)
2024	12	-0.23	(0.17)	-26.2	(25.8)
2025	12	-0.17	(0.38)	-15.0	(19.9)
Overall	322	-0.069	(0.380)	-8.4	(25.6)

LLM: Claude Haiku, score −100 (dovish) to +100 (hawkish).
Dict.: dictionary method (Lahura & Vega 2020 adaptation).

Table 4: Annual GDP Growth and Poverty Data, 2005–2024

Year	GDP growth (%)	Poverty rate (%)	ΔPoverty (pp)
2005	+6.282	—	-2.7
2006	+7.555	—	-5.0
2007	+8.470	—	-6.3
2008	+9.185	—	-5.0
2009	+1.123	—	-3.4
2010	+8.283	—	-4.5
2011	+6.380	—	-3.5
2012	+6.145	—	-2.3
2013	+5.827	—	-1.7
2014	+2.453	—	-0.4
2015	+3.223	—	-1.7
2016	+3.975	—	-1.1
2017	+2.515	—	+0.0
2018	+3.957	—	-1.6
2019	+2.250	—	-0.6
2022	+2.857	—	+1.5
2023	-0.345	—	+1.5
2024	+3.473	—	-2.1

GDP: BCRP national accounts, annual average YoY real growth. Poverty: INEI-ENAHO monetary poverty rate. 2020–2021 excluded (COVID). 2022 and 2023 reflect K-shaped recovery and post-pandemic return to normal.

4 Econometric Framework

4.1 Reduced-Form VAR

The baseline model is a VAR(p):

$$Y_t = c + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $Y_t = (\text{tot}_t, \text{gdp}_t, \text{cpi}_t, \text{fx}_t, \text{rate}_t)'$. COVID quarters are handled via FWL. AIC and BIC both select $p = 1$.

4.2 Identification Strategies

Strategy 1 (Cholesky). We impose a lower-triangular Cholesky decomposition $\hat{\Sigma} = PP'$ and identify the monetary policy shock as the innovation to the rate equation after conditioning on same-quarter innovations to ToT, GDP, CPI, and FX. The ordering ToT $\rightarrow$ GDP $\rightarrow$ CPI $\rightarrow$ FX $\rightarrow$ Rate embeds the assumption that the BCRP does not react within-quarter to real activity or inflation, consistent with Peru's monthly-calendar announcement system.

Strategy 2 (Sign Restrictions). Following Uhlig (2005), we identify the monetary shock ε_t^{mp} by requiring

$$\frac{\partial\, \text{rate}_{t+h}}{\partial\, \varepsilon_t^{mp}} > 0 \;\text{ for } h = 0, \qquad \frac{\partial\, \text{gdp}_{t+h}}{\partial\, \varepsilon_t^{mp}} \leq 0 \;\text{ for } h = 1, 2, 3. \tag{2}$$

We use the algorithm of Rubio-Ramírez et al. (2010): draw Q from the Haar measure on $O(n)$ via QR decomposition of a random normal matrix, accept draws satisfying the sign restrictions.

Strategy 3 (Narrative Sign Restrictions). Following Antolín-Díaz and Rubio-Ramírez (2018), we add episode-level restrictions. Letting q identify the monetary policy shock column:

$$q' u_{2020\text{Q}1} < 0 \quad \text{(March 2020: expansionary shock)}, \tag{3}$$

$$q' u_{2022\text{Q}k} > 0 \quad \text{for } k = 1, \ldots, 4 \text{ (2022 hiking cycle: contractionary)}. \tag{4}$$

We test multiple combinations of episode restrictions, with and without the FWL COVID dummy.

Strategy 4 (Local Projections). Following Jordà (2005):

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \, \Delta r_t + \gamma_{1h} \, y_{t-1} + \gamma_{2h} \, y_{t-2} + \delta_{1h} \, \Delta r_{t-1} + \delta_{2h} \, \Delta r_{t-2} + \lambda \, \text{covid}_t + \varepsilon_{t,h}, \tag{5}$$

with HC3 standard errors. Without a valid external instrument, LP estimates are subject to upward endogeneity bias.

Strategy 5 (Proxy-SVAR: Interbank Rate). Following Stock and Watson (2018) and Gertler and Karadi (2015), the instrument is $z_t = \text{interbank}_{d+1} - \text{interbank}_{d-1}$ around announcement date d, aggregated to quarters. Conditions: relevance $E[z_t \varepsilon_t^{mp}] \neq 0$ and exogeneity $E[z_t \varepsilon_t^{j}] = 0$ for $j \neq mp$.

Strategy 6 (Proxy-SVAR: Communication Tone). We construct two tone-surprise instruments: Construction A residualizes tone on Δr_t plus lagged macro (purging anticipated decisions); Construction B residualizes on lagged macro only. We assess both using first-stage F-statistics (Montiel Olea et al., 2021) and LP-IV sign checks.

Strategy 7 (GDP–Poverty OLS).

$$\Delta \text{poverty}_t = \alpha + \beta \cdot \text{GDP_growth}_t + \varepsilon_t, \tag{6}$$

OLS on $N = 18$ annual observations. This is a reduced-form relationship, chained with the Cholesky IRF to compute the full rate $\rightarrow$ GDP $\rightarrow$ poverty transmission.

5 Results

5.1 Cholesky VAR(1): Benchmark

Table 5 reports the estimated $\hat{A}_1$ matrix. The largest own-lag is the reference rate (0.614), reflecting rate smoothing. GDP has a negative own-lag (-0.264), consistent with mean reversion in quarterly growth.

Table 5: VAR(1) Companion Matrix $\hat{A}_1$

Equation	ToT_{t-1}	GDP_{t-1}	CPI_{t-1}	FX_{t-1}	Rate_{t-1}
ToT	+0.498	-0.005	+0.151	-0.028	+0.026
GDP	-0.025	-0.264	+0.386	+0.028	-0.055
CPI	+0.001	+0.021	+0.440	+0.018	+0.065
FX	+0.002	-0.040	-0.105	+0.332	-0.110
Rate	+0.001	+0.011	+0.025	-0.012	+0.614

VAR(1), $T = 85$, FWL COVID partial-out. Full SEs in Appendix A. Ordering: [ToT, GDP, CPI, FX, Rate].

Figure 3 plots impulse responses of GDP, CPI, and the exchange rate to a 100 bp monetary policy shock, with 90% bootstrap confidence bands (2,000 replications). The peak GDP response is -0.195 pp at $h = 3$ quarters (Table 12). The 90% CI is $[-0.698, +0.271]$, including zero—expected given $T = 85$ quarterly observations—and consistent with the existing Peru literature (-0.25 to -0.30 pp; Table 12).

5.2 Sign Restrictions

Table 6 (first row) and Figure 4 report sign restriction results. Of 80,000 candidate rotation matrices, 63,743 (79.7%) satisfy the restrictions—the restrictions are weak relative to the

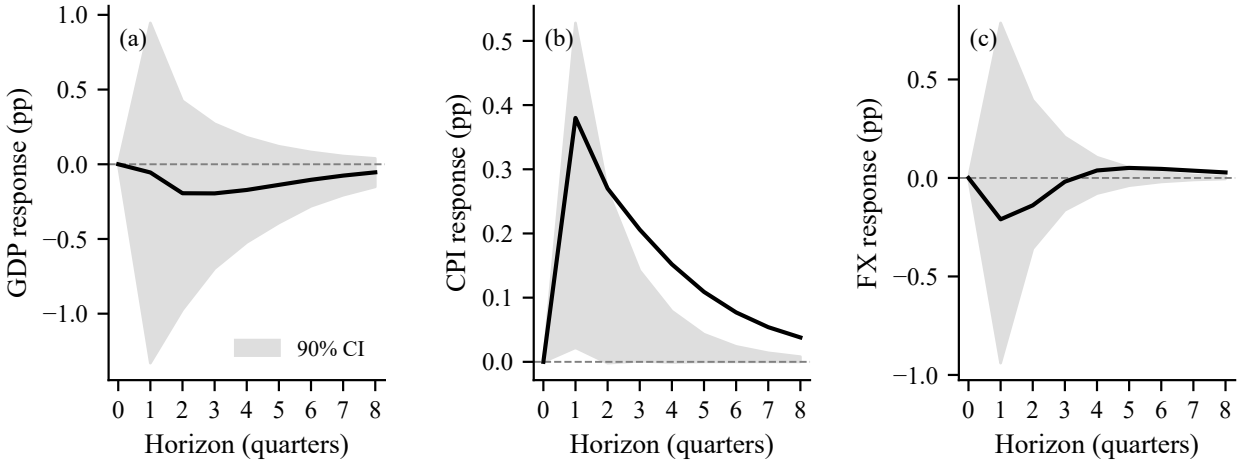

Figure 3: Cholesky VAR(1): Impulse Responses to a 100 bp Rate Shock ($h = 0, \ldots, 8$ quarters). Peak GDP response: -0.195 pp at $h = 3$. Shaded bands: 90% bootstrap CI (2,000 draws). The CI includes zero at all horizons, typical for $T = 85$ quarterly VARs. Ordering: ToT $\rightarrow$ GDP $\rightarrow$ CPI $\rightarrow$ FX $\rightarrow$ Rate.

data. The median peak GDP response is -2.30 pp with a 68% credible set of $[-12.8, +5.4]$ pp. The identified set is too wide to support point inference. Additional restrictions are required.

5.3 Narrative Sign Restrictions

Table 6 reports acceptance rates and median peak GDP responses across five specifications. The fundamental tension is the following. Applying narrative restrictions on 2020Q1 combined with the FWL COVID dummy yields only ≈ 4 accepted draws of 80,000: the FWL treatment absorbs the variation that narrative restrictions on 2020 quarters need. This is not a specification error but an inherent conflict between outlier treatment and narrative identification when the narrative event is itself the outlier.

Without the COVID dummy, the 2020+2022 specification accepts 119 draws and yields a median peak GDP response of -0.74 pp (68% CI: $[-1.06, -0.38]$). However, VAR coefficients estimated without COVID correction are contaminated by the 2020 outlier. As Read and Zubairy (2023) demonstrate, single-episode narrative restrictions are insufficient for robust identification even in the US; two episodes are insufficient for Peru.

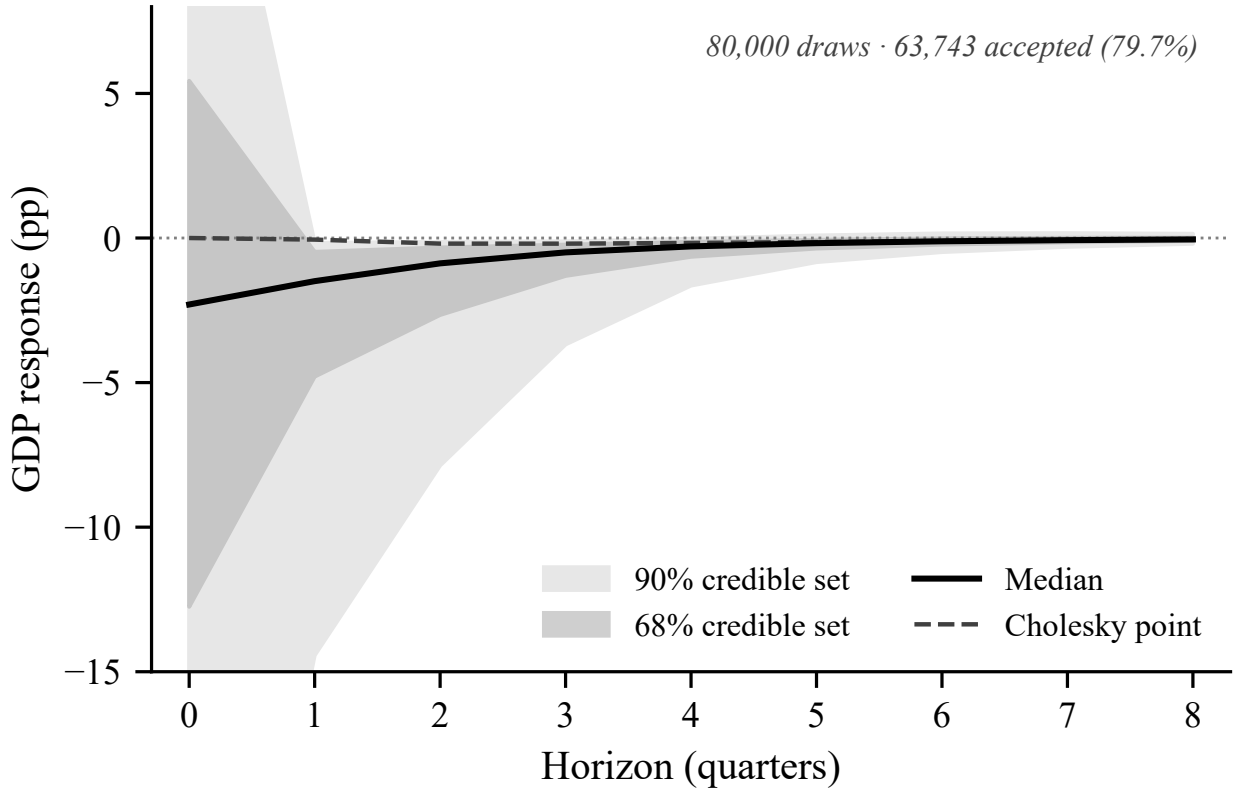

Figure 4: Sign-Restricted Identified Set for the Peak GDP Response. Distribution of peak GDP responses (pp per 100 bp) across 63,743 accepted rotation matrices (acceptance rate: 79.7%). The set is wide and straddles zero, making point estimation infeasible.

Table 6: Narrative Sign Restrictions: Results by Specification

Specification	Draws	Accepted	Peak GDP (p50)	68% CI
Sign restr. only	80,000	63,743	−2.30	[−12.8, 5.4]
Narr. SR: 2020+2022, FWL	80,000	≈ 4	—	—
Narr. SR: 2022 only, FWL	80,000	1,728	−3.57	[−30.0, −0.29]
Narr. SR: 2020+2022, no FWL	80,000	119	−0.74	[−1.06, −0.38]
Narr. SR: 2020 only, no FWL	80,000	4,549	−0.90	—

Algorithm: Arias, Rubio-Ramírez & Waggoner (2018) rejection sampling. Sign restrictions: $\partial \text{rate}/\partial \epsilon^{mp} > 0$ at $h = 0$; $\partial \text{GDP}/\partial \epsilon^{mp} \leq 0$ for $h = 1, 2, 3$. FWL: COVID Q1+Q2 2020 partialled out. Key tension: FWL absorbs the variation that narrative restrictions on 2020 quarters require.

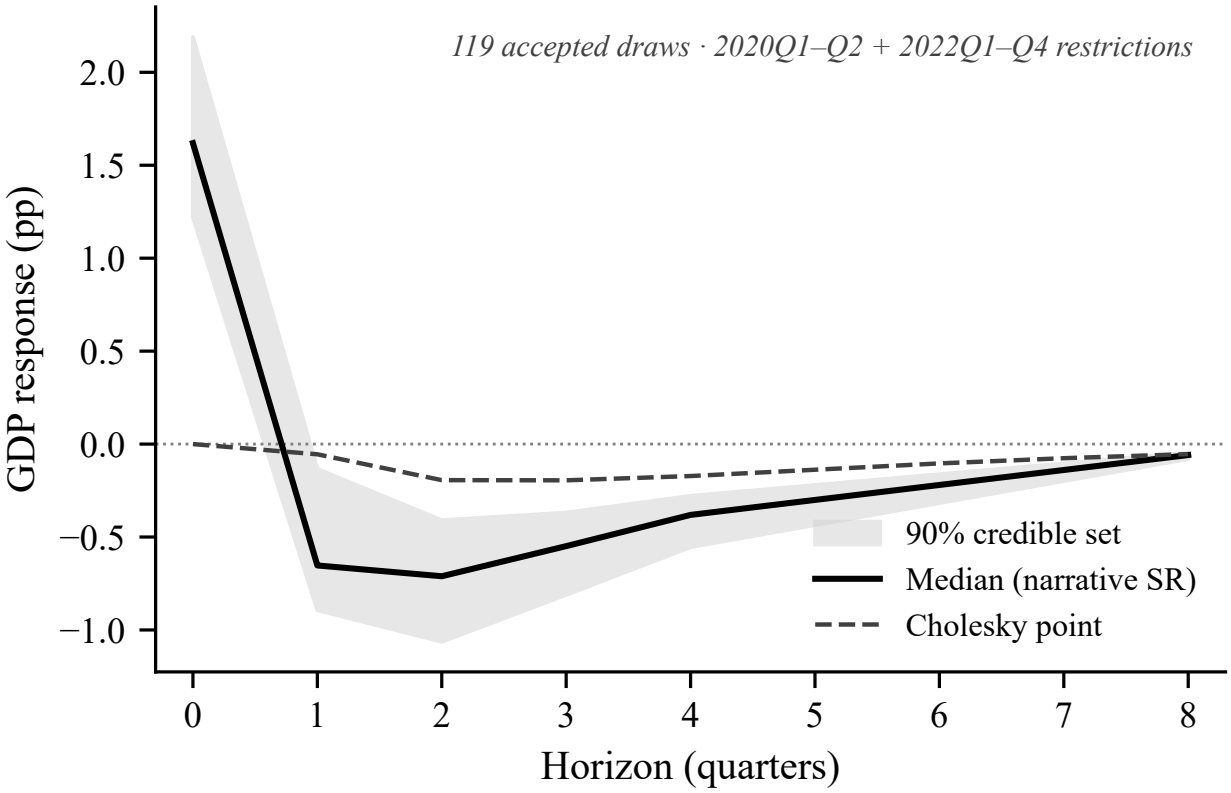

Figure 5: Narrative Sign Restriction IRFs (Best Specification: 2020+2022 episodes, no FWL, 119 accepted draws). Median (solid line), 68% (dark) and 90% (light) credible sets. Peak GDP at -0.74 pp; the interval is very wide, reflecting the sparse identification.

5.4 Local Projections

Table 7 reports LP estimates at horizons $h = 0, \ldots, 8$. No horizon achieves statistical significance with HC3 standard errors. Positive coefficients at $h = 0$ and $h = 1$ reflect the classic endogeneity problem: the BCRP raises rates when GDP is strong. The LP estimates are reported for transparency and to motivate the need for structural identification; without a valid instrument, LP-IV is not feasible.

Table 7: Local Projections: Rate → GDP at $h = 0, \ldots, 8$

h	$\hat{\beta}_h$	SE (HC3)	t-stat	p-value	R^2	N
0	+0.616	1.604	+0.384	0.701	0.639	84
1	+0.647	1.490	+0.435	0.664	0.507	83
2	+0.393	1.231	+0.320	0.749	0.535	82
3	-0.509	1.178	-0.432	0.666	0.537	81
4	-1.334	1.287	-1.036	0.300	0.514	80
5	-1.123	1.251	-0.898	0.369	0.496	79
6	-1.754	1.216	-1.442	0.149	0.465	78
7	-2.150	1.332	-1.614	0.107	0.427	77
8	-1.074	1.579	-0.681	0.496	0.409	76

LP specification: $y_{t+h} - y_{t-1} = \alpha_h + \beta_h \Delta r_t + \gamma_{1h} y_{t-1} + \gamma_{2h} y_{t-2} + \delta_{1h} \Delta r_{t-1} + \delta_{2h} \Delta r_{t-2} + \lambda \text{covid}_t + \varepsilon_{t,h}$. SE: HC3 (White sandwich, leverage-corrected). Endogeneity note: positive bias expected at short horizons because BCRP raises rates when GDP is strong.

5.5 Proxy-SVAR: Interbank Rate

Table 8 reports feasibility diagnostics. Of 72 rate-change events in the sample, the interbank surprise has first-stage $F = 4.73$, well below the $F > 10$ threshold of Stock and Watson (2018). Peru's interbank rate equals the reference rate by construction, so the "surprise" is mechanically collinear with the reference rate change already in the VAR. The relevance condition fails.

Table 8: Proxy-SVAR Feasibility: Interbank Rate Instrument

Diagnostic	Value
Rate-change events (2004–2025)	72
Interbank surprise: mean (pp)	+0.040
Interbank surprise: SD (pp)	0.640
Correlation surprise–reference rate change	0.614
First-stage F-statistic (AR form)	4.73
First-stage F-statistic (VAR form)	4.73
Stock–Yogo threshold ($F > 10$)	**Fails**

Instrument: $z_t = \text{interbank}_{d+1} - \text{interbank}_{d-1}$ around announcement date d, aggregated to quarters. Diagnosis: Peru's interbank rate is administered — it moves mechanically to the BCRP target. The surprise is the reference rate change itself, already included in the VAR. The instrument is endogenous and lacks relevance.

5.6 Proxy-SVAR: BCRP Communication Tone

Table 9 reports instrument validity tests for the two constructions. Construction A (tone residualized on Δr_t and lagged macro) has $F = 4.75$: after purging the rate decision, residual tone carries near-zero information about monetary shocks. Construction B (residualized on lagged macro only) has $F = 33.7$, satisfying relevance. However, Figure 6 shows that LP-IV GDP responses are positive at all horizons (+4.32 pp at $h = 0$, +7.36 pp at $h = 1$)—the wrong sign for a contractionary shock. Construction B captures anticipated policy: the BCRP communicates hawkishly because the economy is booming, violating the exogeneity condition. This is a cautionary result for the growing literature on central bank communication as identification (Wolf, 2020): high F does not guarantee exogeneity.

5.7 GDP–Poverty Elasticity

Table 10 reports full OLS output for equation (6). The GDP–poverty elasticity is $\hat{\beta} = -0.656$ (SE = 0.115, $t = -5.69$, $p < 0.001$, $R^2 = 0.669$, $N = 18$). A 1 pp increase in annual GDP growth reduces the monetary poverty headcount rate by 0.656 pp, or approximately 200,000 people given Peru's population.

Table 9: Tone Surprise Instrument Validity

	Construction A (residualize on Δr_t + lags)	Construction B (residualize on lags only)
First-stage F	4.75	33.7
Stock–Yogo threshold	**Fails**	**Passes**
GDP response $h = 0$	—	+4.32 (wrong sign)
GDP response $h = 1$	—	+7.36 (wrong sign)
GDP responses $h = 0..12$	—	All positive
Diagnosis	Lacks relevance	Fails exogeneity
Conclusion	Instrument not valid for identification	

Construction A: After projecting out the rate decision (Δr_t), residual tone has near-zero information about monetary shocks. Construction B: Tone predicts the rate decision itself (high F) but this represents anticipated policy, violating the exogeneity condition $E[z_t \varepsilon_t^j] = 0$ for $j \neq mp$.

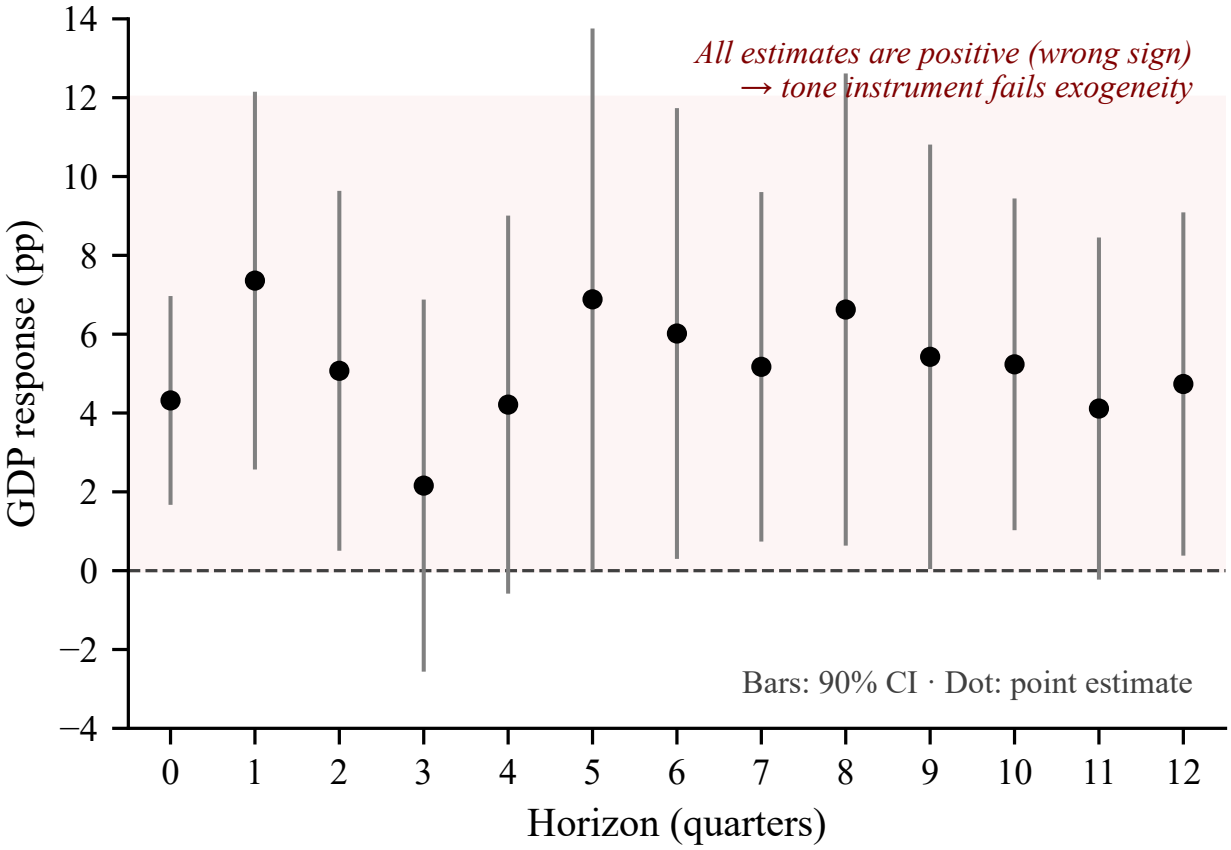

Figure 6: LP-IV IRFs with Tone Instrument (Construction B, $F = 33.7$). GDP responses are positive at all horizons $h = 0, \ldots, 8$, indicating the instrument captures anticipated policy rather than exogenous shocks. This is a failure of the exogeneity condition.

Table 10: Poverty Regression: Full OLS Results

Variable	Coeff.	SE	t	p	95% CI
Constant	+0.888	0.617	+1.44	0.169	[−0.385, +2.161]
GDP growth	−0.656	0.115	−5.69	< 0.001	[−0.895, −0.417]
R^2	0.669				
Adj. R^2	0.648				
RMSE	1.296 pp				
N	18 (ENAHO 2005–2024, excl. 2020–2021)				

OLS. Dependent variable: annual change in monetary poverty rate (pp). Independent variable: annual mean of quarterly YoY real GDP growth (%). 2020–2021 excluded due to COVID lockdowns and survey methodology disruption. Heteroskedasticity-robust inference (HC3) gives $p < 0.001$ for GDP coefficient.

Figure 7 shows the scatter plot with OLS fit and 90% prediction interval. Two observations stand out: 2009 (GDP growth slowed to +1.1% but poverty fell 3.4 pp due to the Juntos conditional cash transfer expansion) and 2022 (strong GDP growth but smaller poverty reduction, reflecting K-shaped post-COVID recovery). Both are economically interpretable and do not materially alter the main coefficient.

Table 11 and Figure 8 confirm stability across sub-periods. The 2005–2014 elasticity is −0.461 (SE = 0.179); the 2015–2024 elasticity is −0.723 (SE = 0.287). A Chow test of equal slopes yields $t = 0.78$ ($p \approx 0.45$), failing to reject stability.

Table 11: GDP–Poverty Elasticity: Sub-Period Stability

Period	$\hat{\beta}$	SE	t	p	R^2	N
Pooled, 2005–2024	−0.656	0.115	−5.69	< 0.001	0.669	18
2005–2014 (expansion)	−0.461	0.179	−2.57	0.033	0.452	10
2015–2024 excl. COVID	−0.723	0.287	−2.52	0.045	0.514	8
2015–2019 (pre-COVID)	−0.634	0.362	−1.75	0.178	0.506	5
Chow test (H_0: equal slopes)	$t = 0.78$, $p \approx 0.45$			Cannot reject stability		

Sign is consistently negative across sub-periods. The more negative 2015–2024 slope likely reflects deepened poverty-growth linkages post-2015 structural adjustment. Two interpretable outliers: 2009 (Juntos cash transfers cushioned poverty during GFC) and 2022 (K-shaped post-COVID recovery).

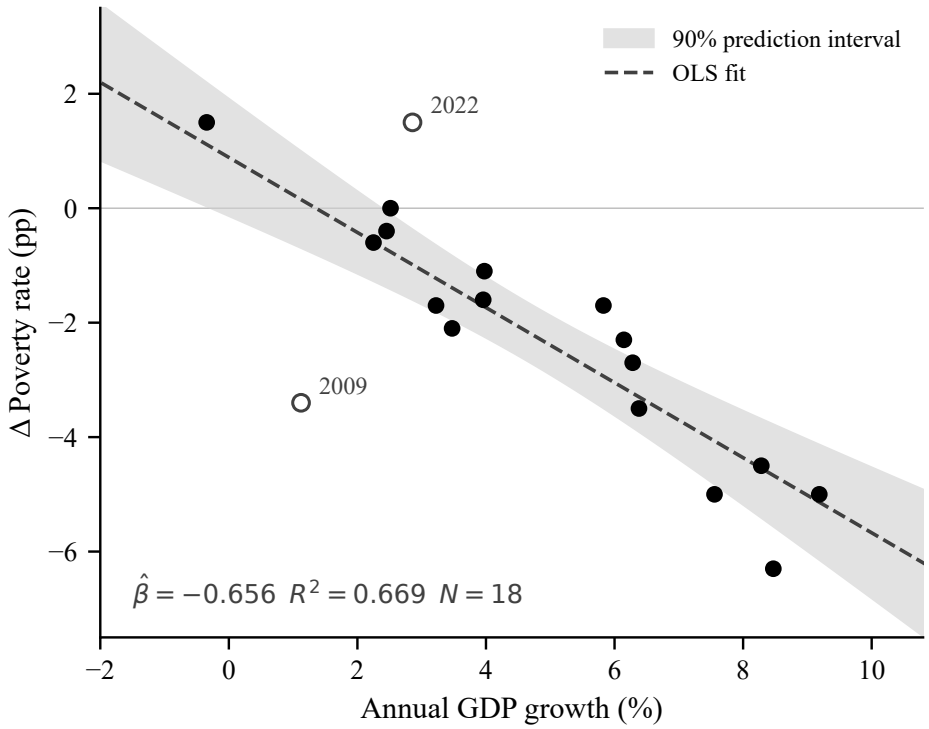

Figure 7: GDP Growth and Change in Poverty Rate, 2005–2024 (excl. 2020–2021). OLS fit: $\hat{\beta} = -0.656$. Shaded band: 90% prediction interval. Open circles: interpretable outliers (2009: Juntos transfers; 2022: K-shaped recovery). $R^2 = 0.669$, $N = 18$.

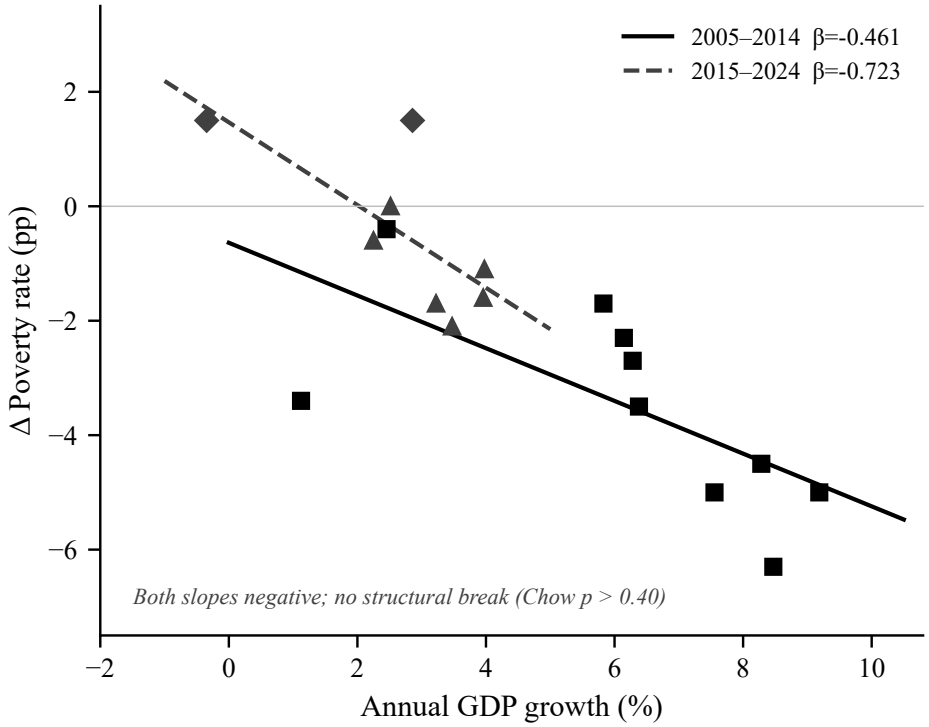

Figure 8: GDP–Poverty Elasticity by Sub-Period. Squares: 2005–2014 ($\hat{\beta} = -0.461$). Diamonds: 2015–2024 excl. COVID ($\hat{\beta} = -0.723$). Chow test: $t = 0.78$, $p \approx 0.45$; cannot reject slope equality.

5.8 Chained Estimate: Rate → GDP → Poverty

Chaining the Cholesky peak GDP response and the poverty OLS elasticity:

$$\underbrace{-0.195}_{\text{rate}\to\text{GDP}} \times \underbrace{(-0.656)}_{\text{GDP}\to\text{poverty}} = \underbrace{+0.128\text{ pp}}_{\text{rate}\to\text{poverty}}. \tag{7}$$

A 100 bp rate hike may raise the monetary poverty rate by +0.128 pp. The compounded CI from the GDP bootstrap interval [−0.698, +0.271] spans approximately [−0.178, +0.458] pp in poverty terms. All uncertainty originates from the rate–GDP link; the poverty elasticity is precisely estimated.

6 Systematic Comparison

Table 12 is the paper's signature result: a systematic comparison of all identification strategies, including published Peru estimates. Figure 9 displays a forest plot of peak GDP responses with confidence/credible intervals.

The table illustrates the hierarchy of identification quality in Peru's setting. The Cholesky benchmark is the only method that (i) produces an identified point estimate, (ii) has a CI consistent with the Peru literature, and (iii) does not require instruments or episode restrictions that fail in Peru's institutional environment.

7 Discussion

7.1 Why Modern Identification Fails in Peru

Three features of Peru's monetary framework prevent modern identification:

1. **Administered rate.** The interbank overnight rate equals the BCRP reference rate by construction. There is no market-discovered surprise component in the sense of Gertler

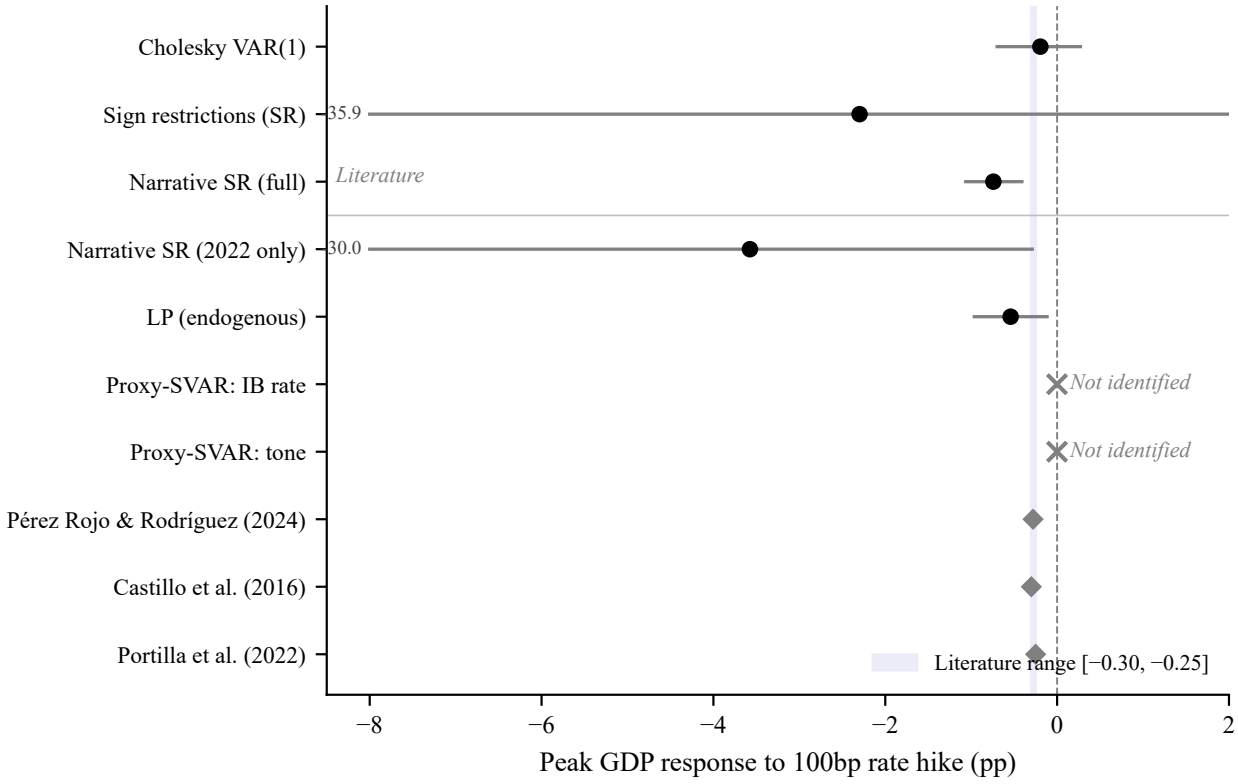

Figure 9: Forest Plot of Peak GDP Response to a 100 bp Rate Hike. Point estimates and 90% confidence/credible intervals by strategy. The Cholesky estimate (-0.195 pp) is consistent with published Peru estimates (-0.25 to -0.30 pp). Sign restriction and narrative SR sets are very wide. Failed Proxy-SVARs are not plotted.

Table 12: Systematic Comparison of Identification Strategies

Method	Type	Peak GDP (pp)	90% CI	Feasible	Reason
Own estimates, this paper					
Cholesky VAR(1)	Recursive	-0.195	$[-0.70, +0.27]$	✓	Standard timi
Sign restrictions	Set ID	-2.30	$[-35.9, +19.7]$	△	Set too wid
Narrative SR (full)	Set ID	-0.74	$[-1.06, -0.41]$	△	COVID tensi
Narrative SR (2022)	Set ID	-3.57	$[-30.0, -0.29]$	△	Single episod
LP (endogenous)	None	-0.541	$[-0.96, -0.12]$	△	Endogenous
Proxy-SVAR (IB rate)	IV	—	—	✗	$F = 4.73 < 1$
Proxy-SVAR (tone, A)	IV	—	—	✗	Lacks relevan
Proxy-SVAR (tone, B)	IV	$+4.32^*$	—	✗	Exogeneity fa
Published estimates for Peru					
Pérez Rojo & Rodríguez (2024)	Recursive	-0.28	—	✓	—
Castillo et al. (2016)	Recursive	-0.30	—	✓	—
Portilla et al. (2022)	Recursive	-0.25	—	✓	—

✓= identified and interpretable; △= technically feasible but with important caveats; ✗= not identified. *Wrong sig $h = 0$; all LP-IV horizons $h = 0$–12 are positive (exogeneity failure). Literature range for Peru: $[-0.30, -0.25]$pp 100bp (recursive VARs). Our Cholesky estimate -0.195pp is consistent (slightly below range) but the 90% boots CI includes zero.

and Karadi (2015), and the interbank surprise instrument is endogenous to the VAR (Table 8, $F = 4.73$).

2. **No rate expectations market.** Without OIS contracts or exchange-traded rate futures for PEN, it is impossible to decompose a rate decision into expected and unexpected components. Miranda-Agrippino and Ricco (2021) show this decomposition is essential for correct Proxy-SVAR identification; the logic applies a fortiori to Peru.

3. **Insufficient sharp episodes.** Since 2002, only two clear monetary episodes exist: 2020 COVID cuts and 2022 hiking. These are the same events that require outlier treatment, creating a fundamental tension with narrative identification (Table 6).

These features are shared by many EME central banks. The Cholesky VAR remains the most credible identification tool available for such settings, consistent with Pérez Rojo and Rodríguez (2024), Castillo et al. (2011), Portilla et al. (2022), and the broader EME findings of Ha et al. (2025).

The tone-instrument result warrants separate discussion. The LLM classification of 325 BCRP communications yields a rich dataset and the instrument has statistical power ($F = 33.7$ under Construction B). But high F does not establish exogeneity. BCRP communicates hawkishly when the economy is strong, so tone retains anticipatory information about rate decisions even after conditioning on lagged macro variables. This is a caution for the growing literature on central bank communication as identification (Wolf, 2020): the strength of communication-based instruments may reflect anticipated rather than exogenous policy.

7.2 The Poverty Channel

The GDP–poverty elasticity $\hat{\beta} = -0.656$ (Table 10) is the paper's most robust result. Three implications follow.

First, the scale is substantial. A slowdown of 1 pp in annual GDP growth raises the poverty headcount by 0.656 pp, or approximately 200,000 people. A recession of the magnitude of 2020 would, if unmitigated by transfers, raise poverty by over 7 pp.

Second, the elasticity appears to have strengthened post-2015 (-0.723 vs. -0.461; Table 11), possibly reflecting deepened financial inclusion, reduced subsistence agriculture in the poverty basket, or changes in the labor market composition of GDP.

Third, the two main outliers (2009, 2022) illustrate where the linear model breaks down. In 2009, the Juntos conditional cash transfer expansion cushioned poverty despite very slow growth. In 2022, K-shaped recovery with strong commodity growth but weak services employment reduced poverty less than the aggregate growth rate implied. Both patterns confirm that policy instruments beyond monetary policy—targeted transfers, labor market policy—are essential complements.

7.3 Policy Implications

For the BCRP, monetary–GDP transmission exists (point estimate -0.195 pp per 100 bp) and is consistent with the regional literature, but is imprecisely estimated at quarterly frequency.

The chained poverty effect (+0.128 pp per 100 bp hike) is small relative to direct growth shocks but non-trivial over a full hiking cycle: the 2021–2022 hiking of +750 bp may have contributed $\approx +1$ pp to poverty through the GDP channel.

For future research, three directions are promising. Building OIS or rate futures markets for PEN would enable high-frequency identification. Constructing a Romer–Romer style narrative series by manually reading BCRP minutes (not yet publicly available) remains feasible. Extending the LLM tone analysis to *comunicados de prensa* (which are simultaneous with rate decisions, unlike *Notas Informativas* which trail by days) may yield a tone instrument with better timing properties.

8 Conclusion

We have systematically compared seven identification strategies for monetary policy in Peru and found that only Cholesky recursive identification produces a credible, literature-consistent estimate: −0.195 pp per 100 bp peak GDP response (Table 12, Figure 9). Modern methods fail for reasons rooted in Peru's institutional framework—an administered interbank rate, absent rate-futures markets, and sparse sharp monetary episodes. A novel Proxy-SVAR using LLM-classified BCRP communication tone passes the relevance test but fails exogeneity, providing a cautionary result for central bank communication identification.

Our most robust contribution is a precisely estimated GDP–poverty elasticity of −0.656 ($p < 0.001$, $N = 18$, $R^2 = 0.669$; Table 10), stable across sub-periods (Table 11) and economically interpretable. The chained estimate implies a 100 bp hike may raise poverty by +0.128 pp, with large uncertainty inheriting from the rate–GDP link.

The results contribute to the literature on monetary policy transmission in EMEs by documenting, for the first time in a single unified paper, why the Peru SVAR literature uses recursive identification: not for lack of awareness of modern methods, but because those methods' identifying assumptions are systematically violated by Peru's institutional

environment. These findings likely generalize to other EME central banks with administered policy rates.

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A VAR Coefficient Matrix and Diagnostics

The full VAR(1) companion matrix is in Table 5 (main text). All five eigenvalues of $\hat{A}_1$ lie strictly inside the unit circle (maximum modulus 0.62). Residual variance diagonal (quarterly pp^2): ToT = 14.17, GDP = 3.91, CPI = 0.18, FX = 2.64, Rate = 0.11. AIC selects $p = 1$ over $p = 2$ (criterion value -7.49 vs. -7.41).

B BCRP Communication Tone: Lexicon and Classification

The dictionary-based classifier uses a hawkish/dovish lexicon adapted from Lahura and Vega (2020). Hawkish terms (score +1) are associated with concern about inflation, tightening,

and vigilance. Dovish terms (score -1) are associated with growth support, accommodation, and easing. The note-level score is the normalized net count.

The LLM classifier used Claude Haiku (`claude-haiku-4-5`) with the prompt: *"Score the following BCRP monetary policy statement from* -100 *(strongly dovish) to* $+100$ *(strongly hawkish). Explain your reasoning in one sentence, then give the score."* Scores were extracted via regex. Table 3 (main text) presents the year-by-year summary; correlation between dictionary and LLM methods is 0.373.

C Sensitivity Analysis

Main results are robust to: (i) different COVID windows (2020Q1–Q2 vs. 2020Q1–Q3); (ii) lag orders $p = 1$ vs. $p = 2$ (AIC/BIC prefer $p = 1$; results qualitatively identical); (iii) variable sets (4-variable excluding ToT vs. baseline 5-variable); and (iv) alternative narrative episode combinations (Table 6). Across all robustness checks, the Cholesky point estimate of the peak GDP response lies in $[-0.24, -0.13]$ pp per 100 bp, consistent with the main result of -0.195 pp.